

SLIDE 1: PROBLEM UNDERSTANDING & OBJECTIVE

Clinical Document Intelligence Hub - Automating Healthcare Document Processing

- ? Problem Context: Healthcare providers spend thousands of hours manually reviewing unstructured clinical documents (discharge summaries, lab reports, intake forms).
- ? Core Bottlenecks: Manual data entry is slow, inconsistent, highly error-prone, and creates administrative delays in patient care pathways.
- ? Project Objective: Build a working proof-of-concept AI prototype that ingests unstructured clinical inputs (PDF, TXT, Image) and surfaces decision-ready structured intelligence.
- ? Value Proposition: Transform fragmented document text into standardized clinical outputs, risk alerts, and evidence-backed next steps ? reducing manual workload by over 70%.

SLIDE 2: SOLUTION ARCHITECTURE & DESIGN FLOW

Hybrid AI Ingestion, Extraction, and Safety Verification Architecture

- ? Document Ingestion Layer: Multi-format parsing accepting PDFs (PyMuPDF), Text files, and Image scans.
- ? Clinical LLM Extraction: Prompt-engineered structured extraction forcing strict Pydantic JSON schema adherence.
- ? Deterministic Safety & Risk Engine: Rules engine post-processes extracted facts (e.g. WBC > 11.0, Penicillin + Amoxicillin allergy contraindications).
- ? Confidence & Evidence Layer: Assigns confidence scores (0-100%) and extracts exact source text evidence quotes for total clinical auditability.
- ? Streamlit Dashboard UI: Clean clinical interface featuring Patient Overview Cards, Risk Flags, Diagnoses/Meds, Lab Tables, and JSON Export.

SLIDE 3: IMPLEMENTATION HIGHLIGHTS

Technical Decisions, AI Logic & System Quality Controls

- ? Structured JSON Schema: Enforced using Pydantic models (Patient, Medication, LabResult, RiskFlag, ClinicalSummary).
- ? Hybrid Reasoning Architecture: LLM used for unstructured text extraction + Deterministic code rules for high-stakes medical thresholds.
- ? Evidence Traceability: Every risk flag and abnormal lab includes exact quote citations from the source document.
- ? Zero-Downtime Multi-Provider Strategy: Seamless support for OpenAI GPT-4o, Google Gemini 2.5, and a zero-dependency deterministic fallback engine.

SLIDE 4: CHALLENGES & LEARNINGS

Addressing Hallucinations, Safety Scoping, and Unstructured Data Variability

- ? Challenge 1 (Hallucination Prevention): Strictly constrained LLM prompt rules to extract ONLY explicitly stated facts; unmentioned fields return null.
- ? Challenge 2 (Clinical Safety): Scoped all recommendations as decision-support suggestions rather than autonomous diagnoses.
- ? Challenge 3 (Document Formatting Variance): Standardized multi-format text extraction pipeline to handle noisy PDFs and plain text notes uniformly.
- ? Key Takeaway: Combining LLM semantic flexibility with deterministic rule validation provides the ideal balance of accuracy and safety in healthcare software.

SLIDE 5: DEMO SUMMARY & NEXT STEPS

Working Prototype Deliverables & Strategic Roadmap

- ? Delivered Artifacts: Full working Streamlit application, sample datasets (PDF/TXT), 1-page README, and 3-minute video demo script.
- ? Next Steps - Healthcare Integration: HL7 / FHIR API export for direct EHR integration (Epic, Cerner, AthenaHealth).
- ? Next Steps - Safety & Security: Enterprise HIPAA compliance, PHI de-identification (PII/PHI anonymization pipeline), and OAuth2 RBAC.
- ? Next Steps - Clinical Workflow: Human-in-the-loop clinician sign-off workflow and multi-document longitudinal patient timelines.